EXPERIMENT 2

Aim:

1. Prepare a program for changing the image to HSV format.
2. Write the python programs to crop an image.
3. Write the program for blurring an image using kernel of shape.
4. Prepare a program to find the contours in an image.

Code:

```
import cv2
import numpy as np
from PIL import Image

path = '/Users/vishrutgrover/coding/sem6/cv/lab2/'
img = cv2.imread(path + 'claw.png')

# 1. HSV conversion
cv2.imwrite(path + 'hsvclaw.png', cv2.cvtColor(img, cv2.COLOR_BGR2HSV))

# 2. Crop image
Image.open(path + 'claw.png').crop((300, 100, 600, 400)).save(path + 'cropclaw.png')

# 3. Blur with kernel
cv2.imwrite(path + 'blurredclaw.png', cv2.filter2D(img, -1, np.ones((5, 5), np.float32) / 25))

# 4. Find contours
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
contours, _ = cv2.findContours(cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)[1],
cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE)
cv2.drawContours(img, contours, -1, (0, 255, 0), 2)
cv2.imwrite(path + 'contoursclaw.png', img)
```

Output:

OpenClaw
THE AI THAT ACTUALLY DOES THINGS.

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